Exercise Sheet 2 – Solution

Even though this PVK is structured into thematic sections and some content has not yet been covered, it is explicitly permitted and *should* be the case that in the problem sets *all the material* covered in the semester and in the lecture may be used.

Exercise 1
Calculate the value of the series

$$\sum_{n=1}^{\infty} \frac{\log(2^n)}{2^n}.$$

Solution: We recognize

$$\sum_{n=1}^{\infty} \frac{\log(2^n)}{2^n} = \log(2) \sum_{n=1}^{\infty} \frac{n}{2^n} = \log(2) \sum_{n=1}^{\infty} n \left(\frac{1}{2}\right)^n.$$

We derive a formula for series of the form $S = \sum_{n=1}^{\infty} nq^n$. We obtain by considering qS

$$qS = \sum_{n=1}^{\infty} nq^{n+1} = \sum_{n=0}^{\infty} nq^{n+1} = \sum_{n=1}^{\infty} (n-1)q^n = \sum_{n=1}^{\infty} nq^n - \sum_{n=1}^{\infty} q^n = S - \left(\frac{1}{1-q} - 1\right) = S - \left(\frac{q}{1-q}\right).$$

Hence we have

$$S(q-1) = -\frac{1}{1-q} = \frac{q}{q-1} \iff S = \frac{q}{(q-1)^2} \quad \text{if } |q| < 1.$$

Thus for our series we have

$$\log(2) \sum_{n=1}^{\infty} n \left(\frac{1}{2}\right)^n = \log(2) \frac{\frac{1}{2}}{(\frac{1}{2}-1)^2} = 2\log(2).$$

Exercise 2 (HS2022 G. Felder)
Calculate the value of the series

$$\sum_{n=1}^{\infty} \frac{1}{n2^n}.$$

Hint: You can differentiate the series to derive a general formula for $\sum_{n=1}^{\infty} \frac{1}{n} q^n$. Note that there are multiple ways of solving this exercise!

Solution: We can derive a general formula for series of the form $S = \sum_{n=1}^{\infty} \frac{1}{n} q^n$ with $|q| < 1$.

$$\frac{dS}{dq} = \frac{d}{dq} \sum_{n=1}^{\infty} \frac{1}{n} q^n = \sum_{n=1}^{\infty} \frac{n}{n} q^{n-1} = \sum_{n=1}^{\infty} q^{n-1} = \sum_{n=0}^{\infty} q^n = \frac{1}{1-q}.$$

We obtain a closed formula for S by integrating with respect to q.

$$S = \int \frac{1}{1-q}\, dq = -\log(1-q) + C,$$

with the condition that $S(q=0)=0$. It follows that $C=0$. The value of our series follows with $q=\frac{1}{2}$.

$$\sum_{n=1}^{\infty}\frac{1}{n2^n}=-\log\left(1-\frac{1}{2}\right)=-\log\left(\frac{1}{2}\right)=\log(2).$$

It is worth noting, that we essentially derived the Taylor series for $\log(1-x)$.

Exercise 3 (HS2022 G. Felder)
Calculate the radius of convergence of the power series

$$\sum_{n=0}^{\infty}\frac{(3n)!}{(n!)^3}x^n.$$

Solution: We use the ratio test (this is often helpful with factorials).

$$\lim_{n\to\infty}\frac{3n!}{n!^3}\frac{(n+1)!^3}{(3(n+1))!}=\lim_{n\to\infty}\frac{3n(3n-1)(3n-2)\cdots}{(n(n-1)(n-2)\cdots)^3}\frac{((n+1)n(n-1)(n-2)\cdots)^3}{3(n+1)(3(n+1)-1)(3(n+1)-2)\cdots}$$

$$=\lim_{n\to\infty}(n+1)^3\frac{3n(3n-1)(3n-2)\cdots}{(3n+3)(3n+2)(3n+1)(3n)(3n-1)\cdots}=\lim_{n\to\infty}\frac{(n+1)^3}{(3n+3)(3n+2)(3n+1)}$$

$$=\lim_{n\to\infty}\frac{n^3+\cdots}{3^3n^3+\cdots}=\frac{1}{27}.$$

Exercise 4 (FS2024, A. Figalli)
Calculate the radius of convergence of the power series

$$\sum_{n=1}^{\infty}\frac{n6^n}{3^{2n+1}+n2^n}x^n.$$

Solution: We use the Cauchy root test

$$\rho=\lim_{n\to\infty}\sqrt[n]{\frac{n6^n}{3^{2n+1}+n2^n}}\sim\lim_{n\to\infty}\sqrt[n]{\frac{n6^n}{3^{2n}}}\sim\frac{6}{3^2}=\frac{6}{9}=\frac{2}{3}.$$

Thus the radius of convergence is $R=\frac{1}{\rho}=\frac{3}{2}$.

Exercise 5 (FS2025, L. Kobel-Keller)
The series

$$\sum_{n=1}^{\infty}\frac{n^n(x-2)^n}{(2n)!}$$

converges...

(A) only for $x=0$.

(B) for all $x\in\mathbb{R}$.

(C) only for $x\in(-1,1)$.

(D) only for $x\in(0,4)$.

Solution: (B) is correct. We compute using the ratio test

$$\frac{a_{n+1}}{a_n} = \frac{\frac{(n+1)^{n+1}}{(2n+2)!}}{\frac{n^n}{(2n)!}} = \frac{(n+1)^{n+1}}{(2n+2)!} \cdot \frac{(2n)!}{n^n} = \frac{(n+1)^{n+1}}{n^n} \cdot \frac{(2n)!}{(2n+2)!}$$
$$= \frac{(n+1)^{n+1}}{n^n} \cdot \frac{1}{(2n+1)(2n+2)} = \left(1+\frac{1}{n}\right)^n \cdot \frac{(n+1)}{(2n+1)(2n+2)}$$

As $n \to \infty$, $\left(1+\frac{1}{n}\right)^n \to e$ and $\frac{(n+1)}{(2n+1)(2n+2)} \to 0$. Thus $R = +\infty$.

Exercise 6 (HS2022 G. FELDER)
Calculate the maximum $m = \max\{3x^3 - x \mid x \in [-1,1]\}$.

Solution: We look for local maxima in $(-1,1)$ and then check the boundaries.

$$f'(x) = 9x^2 - 1 = 0. \iff x = \pm\frac{1}{3}.$$

We substitute and see

$$f\left(-\frac{1}{3}\right) = -\frac{3}{27} + \frac{1}{3} = \frac{2}{9}, \quad f\left(\frac{1}{3}\right) = \frac{3}{27} - \frac{1}{3} = -\frac{2}{9}.$$

At the boundaries we have

$$f(-1) = -3 + 1 = -2, \quad f(1) = 3 - 1 = 2.$$

Thus the maximum is $m = 2$.

Exercise 7 (FS2023 G. FELDER)
Calculate the minimum $m = \min\{x\log(x) \mid x > 0\}$.

Solution: We look for local maxima in $(0,\infty)$ and then check the boundaries.

$$f'(x) = \log(x) + 1 = 0 \iff x = \frac{1}{e}$$

Thus

$$f\left(\frac{1}{e}\right) = \frac{1}{e}\log\left(\frac{1}{e}\right) = -\frac{1}{e}.$$

Since both $f(x) \to 0$ as $x \to 0$, and $f(x) \to \infty$ as $x \to \infty$, the minimum is at $-\frac{1}{e}$.

Exercise 8
Let $D, E \subset \mathbb{R}$ be subsets and $x_0 \in D$ an accumulation point of $D\backslash\{x_0\}$. Let $f : D \to E$ be differentiable at x_0, such that $y_0 = f(x_0)$ is an accumulation point of $E\backslash\{y_0\}$ and $g : E \to \mathbb{R}$ differentiable at y_0. Show that $g \circ f : D \to \mathbb{R}$ is differentiable at x_0 and we have

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0).$$

Solution: We can write the function $g(y)$ as:

$$g(y) = g(y_0) + g'(y_0)(y - y_0) + \omega(y)(y - y_0)$$

$$\omega(y) = \begin{cases} \frac{g(y)-g(y_0)}{y-y_0} - g'(y_0) & \text{if } y \in E \setminus \{x_0\} \\ 0 & \text{if } y = y_0 \end{cases}$$

Because the function g is differentiable at y_0, ω is continuous. Consider the limit $y \to y_0$, the left term then becomes the difference quotient. The differentiability of g ensures that the limit exists.

Set $y = f(x)$ and note that $y_0 = f(x_0)$:

$$g(f(x)) = g(f(x_0)) + g'(f(x_0))[f(x) - f(x_0)] + \omega(f(x))[f(x) - f(x_0)]$$

From this equation we "build"the difference quotient. We subtract $g(f(x_0))$ and divide by $x - x_0$

$$\frac{g(f(x)) - g(f(x_0))}{x - x_0} = g'(f(x_0))\frac{f(x) - f(x_0)}{x - x_0} + \omega(f(x))\frac{f(x) - f(x_0)}{x - x_0}$$

More precisely, for $\omega(f(x)$ due to the differentiability of g at y_0:

$$\lim_{x \to x_0} \omega(f(x)) = \lim_{x \to x_0} \left(\underbrace{\frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)}}_{\to g'(f(x_0))} - g'(f(x_0)) \right) = 0$$

Therefore, for the limit as $x \to x_0$:

$$\lim_{x \to x_0} \left(\frac{g(f(x)) - g(f(x_0))}{x - x_0} \right) = \lim_{x \to x_0} \left(g'(f(x_0)) \underbrace{\frac{f(x) - f(x_0)}{x - x_0}}_{\to f'(x_0)} + \underbrace{\omega(f(x))}_{\to 0} \frac{f(x) - f(x_0)}{x - x_0} \right)$$
$$= g'(f(x_0))f'(x_0)$$

Exercise 9
Let $D, E \subseteq \mathbb{R}$ and let $f : D \to E$ be a continuous, bijective function whose inverse $f^{-1} : E \to D$ is also continuous. Let $\bar{x} \in D$ be an accumulation point of $D \setminus \{\bar{x}\}$ and let f be differentiable at $\bar{x}$ with $f'(\bar{x}) \neq 0$. Prove, that f^{-1} is differentiable at $\bar{y} := f(\bar{x})$ and we have

$$(f^{-1})'(\bar{y}) = \frac{1}{f'(\bar{x})}.$$

Solution: To compute $(f^{-1})'(\bar{y})$, take an arbitrary sequence $(y_n)_{n=0}^{\infty} \subset E \setminus \{\bar{y}\}$ such that $y_n \to \bar{y}$ and set $x_n = f^{-1}(y_n)$. Then

$$\frac{f^{-1}(y_n) - f^{-1}(\bar{y})}{y_n - \bar{y}} = \frac{x_n - \bar{x}}{f(x_n) - f(\bar{x})} = \left(\frac{f(x_n) - f(\bar{x})}{x_n - \bar{x}} \right)^{-1}.$$

By the continuity of f^{-1}, we have $x_n \to \bar{x} = f^{-1}(\bar{y})$. Because f is differentiable at $\bar{x}$ and $f'(\bar{x}) \neq 0$, the difference quotient for f converges to $f'(\bar{x})$. Therefore, the reciprocal of this difference quotient converges to the reciprocal of $f'(\bar{x})$. That is,

$$\left(\frac{f(x_n) - f(\bar{x})}{x_n - \bar{x}}\right)^{-1} \longrightarrow \frac{1}{f'(\bar{x})}.$$

Consequently,

$$\lim_{n\to\infty} \frac{f^{-1}(y_n) - f^{-1}(\bar{y})}{y_n - \bar{y}} = \frac{1}{f'(\bar{x})}.$$

Since the sequence $(y_n)_{n=0}^{\infty}$ was chosen arbitrarily, this implies that the limit of the difference quotient for f^{-1} exists as y approaches $\bar{y}$. Hence,

$$\lim_{y\to\bar{y}} \frac{f^{-1}(y) - f^{-1}(\bar{y})}{y - \bar{y}} = \frac{1}{f'(\bar{x})},$$

which is exactly $(f^{-1})'(\bar{y})$.

Exercise 10
Let $f : [a, b] \to \mathbb{R}$ be a continuous function and differentiable on (a, b). Furthermore, assume $f(a) < 0 < f(b)$. Show that if $f'(x) \neq 0$ for all $x \in [a, b]$, then f has exactly one zero.

Solution: We know that f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$. Thus by the Intermediate Value Theorem there is at least one zero. Assume for contradiction that there is a second zero. This must lie in (a, b) because $f(a) < 0 < f(b)$. Since f is differentiable on (a, b), we can apply the Mean Value Theorem. This implies a vanishing derivative between the two zeros, because there $f(x_1) = f(x_2) = 0$ with $x_1 \neq x_2$ holds. This contradicts the assumption $f'(x) \neq 0$ for all $x \in [a, b]$.